

Gastrointestinal physiology

Digestive Organs

Integrated Biomedical and Clinical Sciences OMFS 512

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Learning Objectives

1. Describe the components of the gastrointestinal system, its blood supply and neural innervation.
2. Describe regulatory mechanism in the GI system
3. Describe phases of digestion
4. Compare and contrast major gastrointestinal secretions.
5. Describe stomach secretion, its regulation and disorders

Gastrointestinal Tract

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graph TD; A[Gastrointestinal Tract] --> B[Primary Digestive Organs]; A --> C[Accessory digestive Organs]; B --> D["Oral cavity, esophagus, stomach, small and large intestine"]; C --> E["Teeth, tongue, salivary glands, exocrine pancreas, liver and gall bladder."];
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Primary Digestive Organs

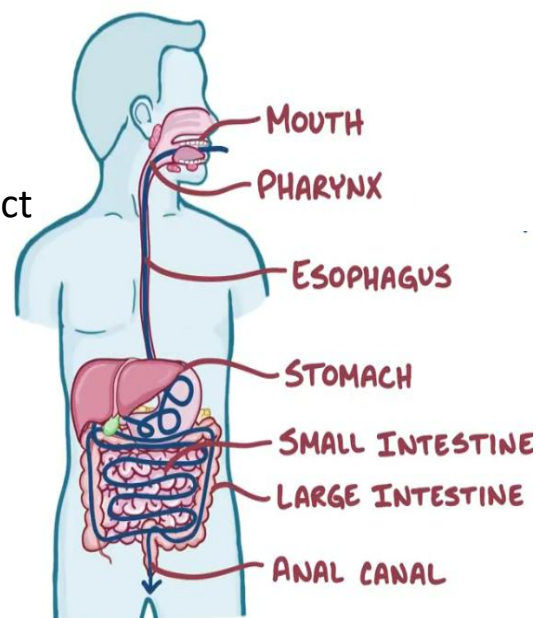
“Oral cavity,
esophagus,
stomach,
small and large
intestine

Accessory digestive Organs

“Teeth, tongue,
salivary glands,
exocrine pancreas,
liver and gall
bladder.

Outline

- Gastrointestinal tract.
- Regulatory mechanisms in the GI tract
- Cephalic phase of digestion.
- Gastric phase of digestion.
- Gastritis and Peptic ulcer disease.

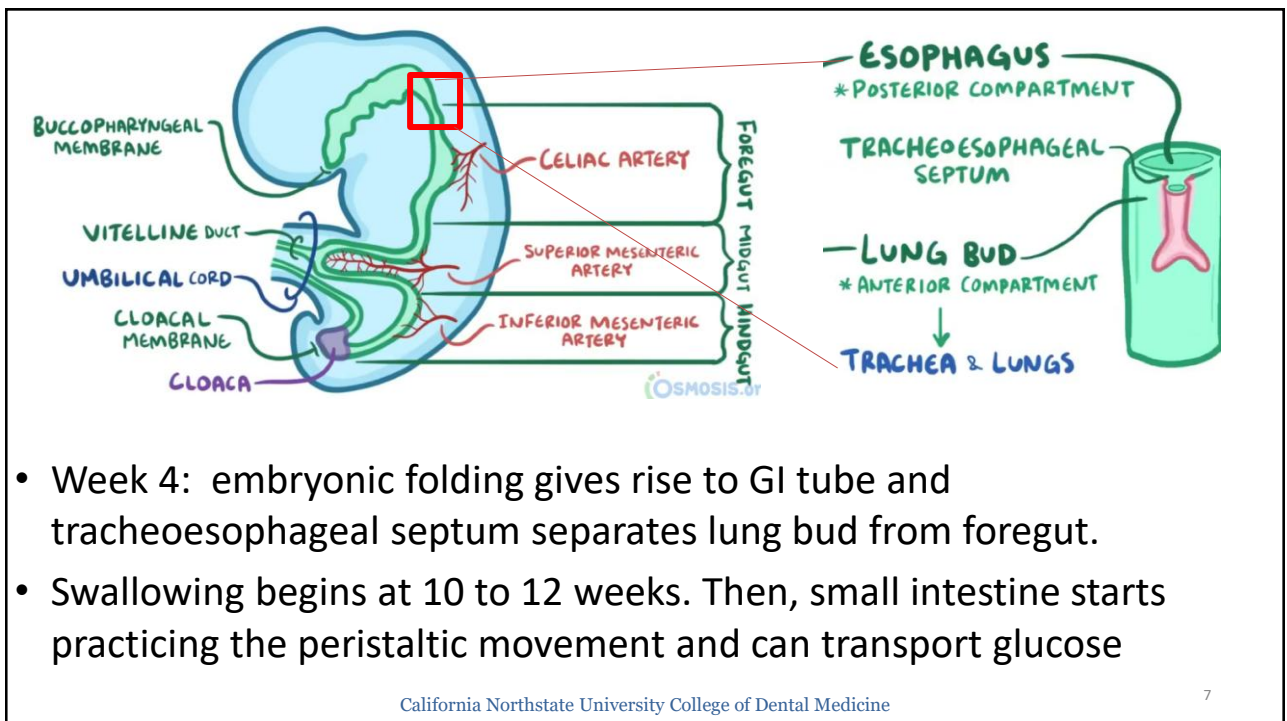


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Embryonic Development

- GI tract develops from the three germ layers.
 - Endoderm layer: epithelial lining and mucosal glands.
 - Mesoderm layer: connective tissue and smooth muscles and remaining structure.
 - Ectoderm: create the enteric nervous system from migrating neural crest cells.

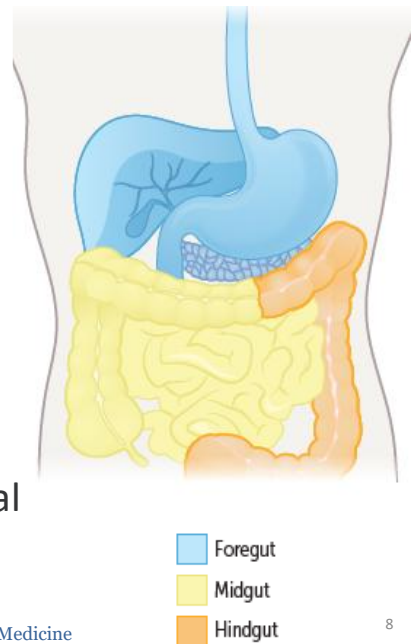


- Week 4: embryonic folding gives rise to GI tube and tracheoesophageal septum separates lung bud from foregut.
- Swallowing begins at 10 to 12 weeks. Then, small intestine starts practicing the peristaltic movement and can transport glucose

1- Foregut, supplied by the celiac artery, formed from the cephalic folding. Includes pharynx to proximal duodenum, liver, pancreas, and biliary tree.

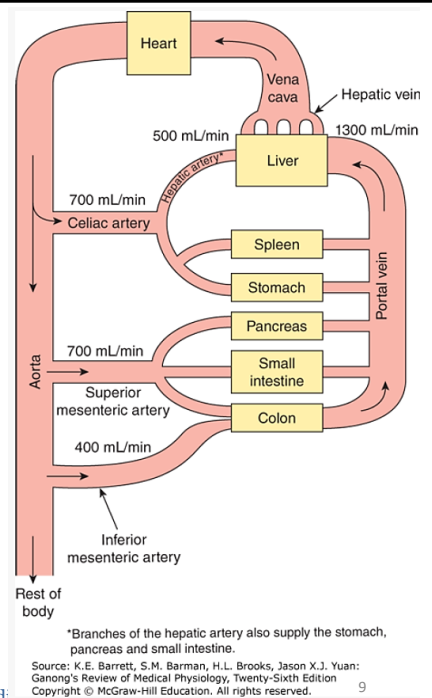
2- Midgut, supplied by the superior mesenteric artery, formed from the lateral folding. Includes distal duodenum to cecum, appendix, and the right colon.

3- The hindgut, supplied by the inferior mesenteric artery and formed from the caudal folding. Develops into the left colon, rectum.



Blood Supply

- Oxygenated blood is delivered via the celiac and mesenteric arteries to various GI organs.
- Blood from the stomach, intestines, pancreas, and spleen drains via the hepatic portal vein to the liver.



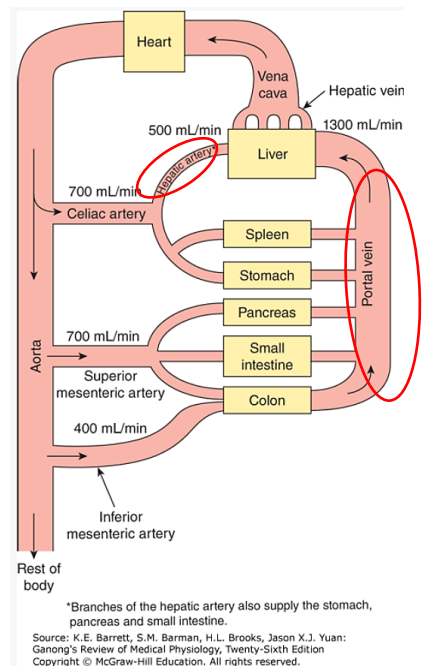
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Liver supply

Dual blood supply (25 % of CO):

- a) 2/3 deoxygenated nutrient-rich supply from the hepatic portal vein.
- b) 1/3, oxygenated blood from the hepatic artery.

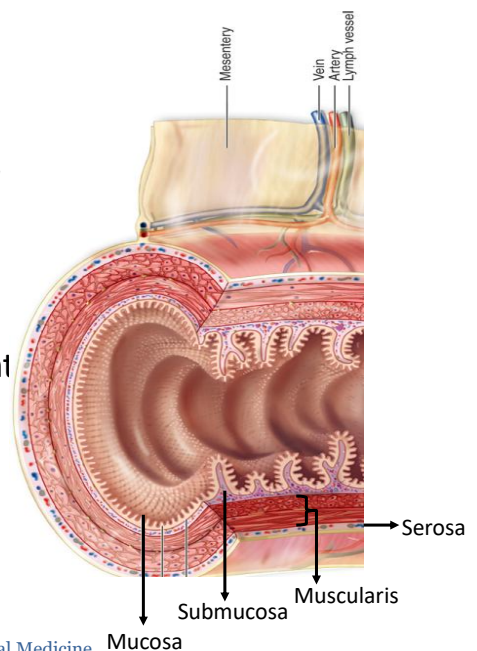
Blood from the portal vein and hepatic artery are mixed in the hepatic sinusoids & drain into vena cava



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General Histology of GI tract

- GI tract is a hollow tube with a variable lumen diameter. Four main layers (mucosa, submucosa, muscularis and serosa).
- Stomach, small intestine, and large intestine are suspended by mesenteries, through which the nerves, blood vessels, and lymphatics are present and encased by the peritoneum.
- Peritoneum is derived from mesoderm (mesothelial cells +thin fibrous tissue), lines the abdominal cavity.



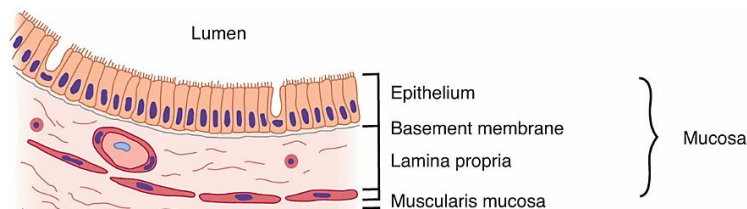
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Mucosa

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Mucosa includes:

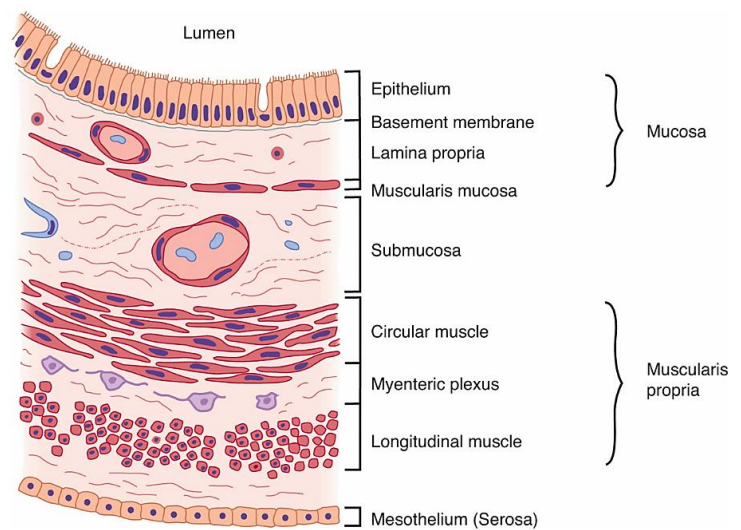
- a) A layer of epithelial cells. Form a continuous lining to the GI tract. First barrier to permit nutrients absorption. Epithelial cell layer adapt in different parts of the GI tract.
- b) Lamina propria: loose connective tissue layer. Occupied by blood vessels, immune and inflammatory cells.
- c) Muscularis mucosa, thin muscle layer. Contraction controls the surface area of the mucosa.



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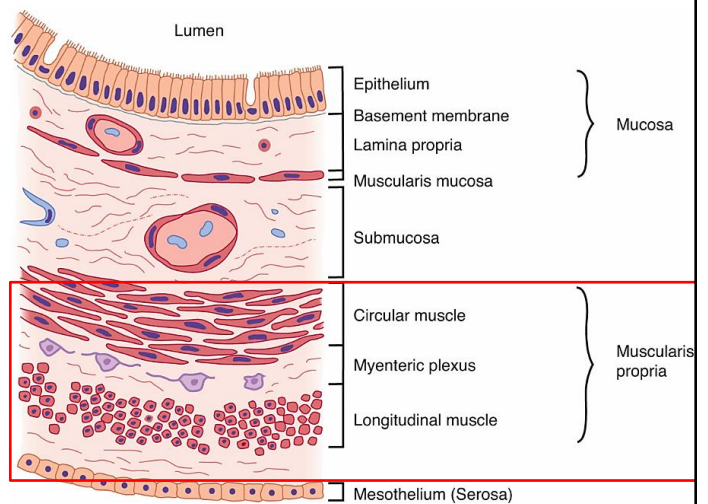
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Submucosa: denser connective tissue that connects the mucosa to the muscularis layer. It includes the submucosal nerve plexus (Meissner), larger blood and lymphatic vessels



Muscularis propria (aka muscularis externa). Includes the myenteric nerve plexus, circular and longitudinal muscles.

- Circular muscle controls the diameter of the GI tract (i.e contraction of this muscle causes a decrease in diameter of the GI lumen).

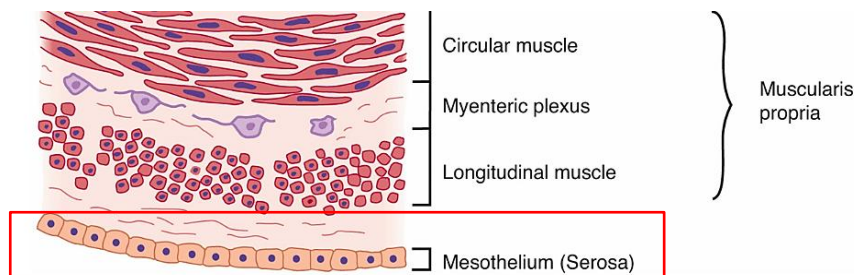


- The longitudinal muscle controls the GI length (i.e, its contraction causes shortening of the GI segment). Coordinated action of both muscle layers lead to peristalsis.

Serosa or adventitia layer. Outermost thin connective tissue layer that encloses the GI organs.

Adventitia is mainly composed of connective tissue (example, esophagus) .

Serosa contain connective tissue and simple squamous epithelium (mesothelium) with secretory cells that secrete viscous lubricant fluid that protects the organs and reduces friction (example stomach and intestine).

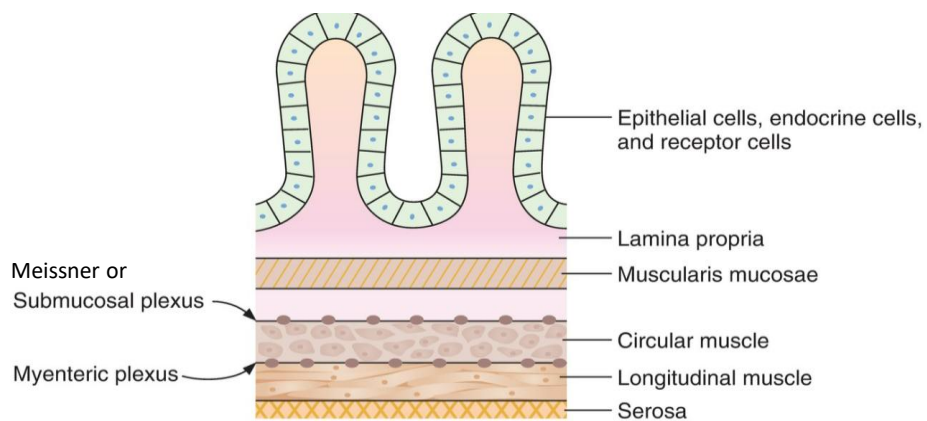


Neural innervation

Intrinsic and extrinsic

A) Intrinsic innervation via the enteric nervous system

- Relays signals to the CNS via the autonomic nervous system. Can also function independently.
- Controls most functions of the GI tract, especially motility and secretion, even in the absence of extrinsic innervation.
- Neurotransmitters of the enteric nervous system include acetylcholine, norepinephrine, serotonin, GABA.



- **Composed of**

- Meissner (submucosal) plexus controls secretion and blood flow. It receives sensory information from chemoreceptors and mechanoreceptors in the GI tract.
- Myenteric plexus mainly controls the motility of the GI smooth muscles.

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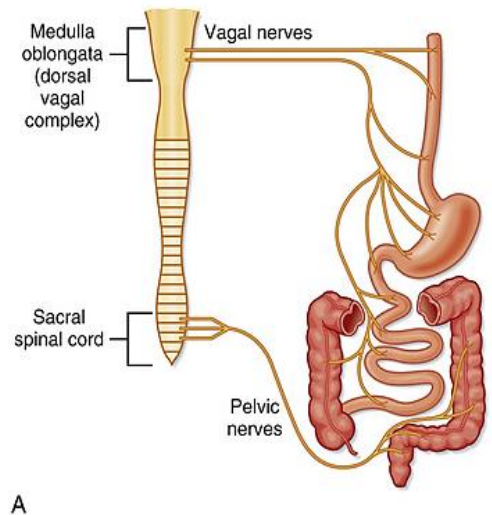
B) Extrinsic innervation

Afferent fibers carry sensory information from GI chemoreceptors and mechanoreceptors to the CNS.

CNS modulate the GI function via:

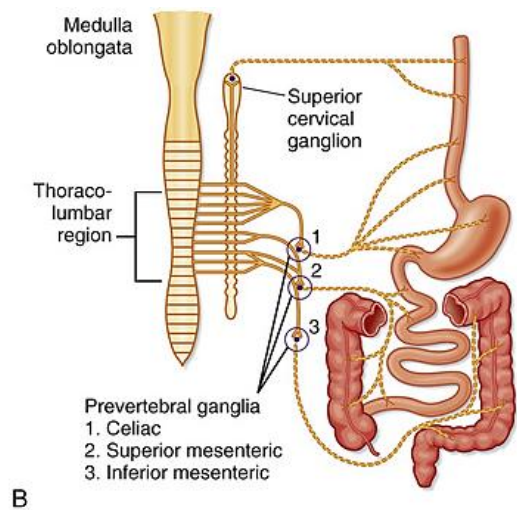
➤ Parasympathetic nervous system

Via the vagus (foregut and midgut) and pelvic splanchnic (to hindgut) nerves. Exerts excitatory influence. Main neurotransmitter: Ach.



➤ Sympathetic nervous system

SNS originate from the thoracic and upper lumbar region in the spinal cord (between T8 and L2). Exerts inhibitory influence on the GI functions via inhibiting the enteric nervous system. Main neurotransmitter: NE.



Receptors in the GI tract

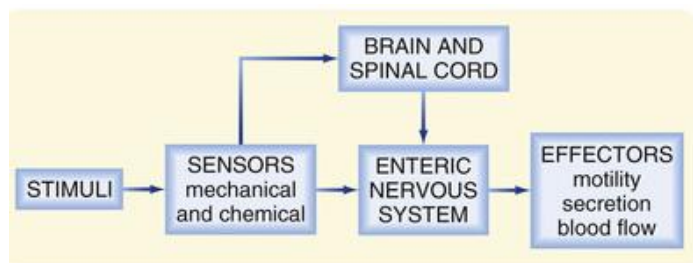
Mechanoreceptors

Respond to stretch, presence of food in the GI tract.

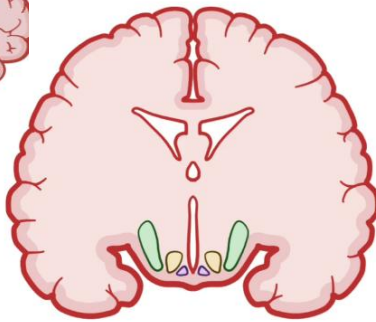
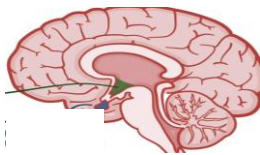
Chemoreceptors

Respond to changes in pH, chemicals in food or end-products of digestion.

Collectively, those receptors send sensory signals and trigger reflexes that can activate or inhibit digestion.



Hypothalamic control



LATERAL HYPOTHALAMIC AREA

↳ FEEDING CENTER



SENSATION of **HUNGER**

ARCULATE NUCLEUS

↳ SWITCHBOARD

↳ RECEIVES SIGNALS from the **GI TRACT**

↳ REGULATES the **FEEDING CENTER** & **SATIETY CENTER**

VENTROMEDIAL NUCLEI

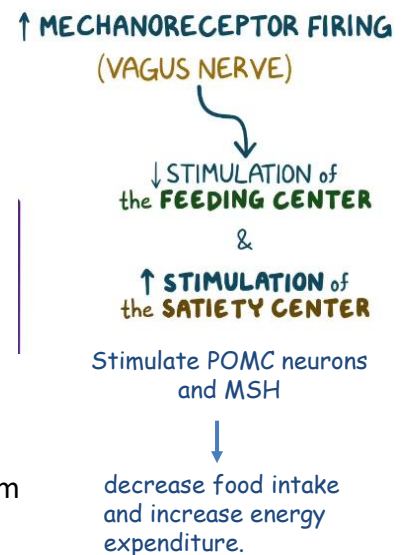
↳ SATIETY CENTER



SENSATION of **FULLNESS**

Satiety center, in ventromedial nucleus of hypothalamus to decrease appetite.

- Stretch mechanoreceptors (via vagus) modulate arcuate nucleus. Activate anorexigenic neurons, proopiomelanocortin (POMC) neurons & α -melanocyte–stimulating hormone in hypothalamic centers = decreased appetite.
- Regulating hormones (stimulate satiety and inhibit hunger):
 - Long term: Leptin, secreted by fat cells. It stimulates anorexigenic neurons and inhibits orexigenic neurons (decrease appetite).
 - Short-term: Insulin (pancreas), peptide YY and GLP-1 (from L cells in intestine).

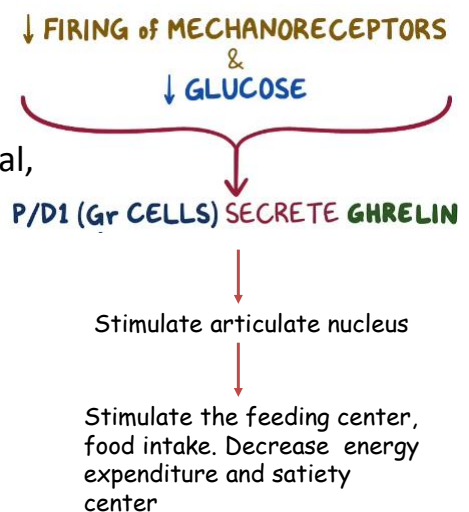


- **Feeding center**, lateral hypothalamic area, stimulates appetite.

- Mechanoreceptors detect empty stomach (decreased stimulation) & suppress vagal signal, modulate arcuate nucleus.

- Low blood glucose stimulates ghrelin secretion and further stimulates the arcuate nucleus.

- Arcuate nucleus stimulates orexigenic neurons to release neuropeptide Y (hypothalamic centers) = stimulate appetite & inhibit anorexigenic neurons/satiety center.

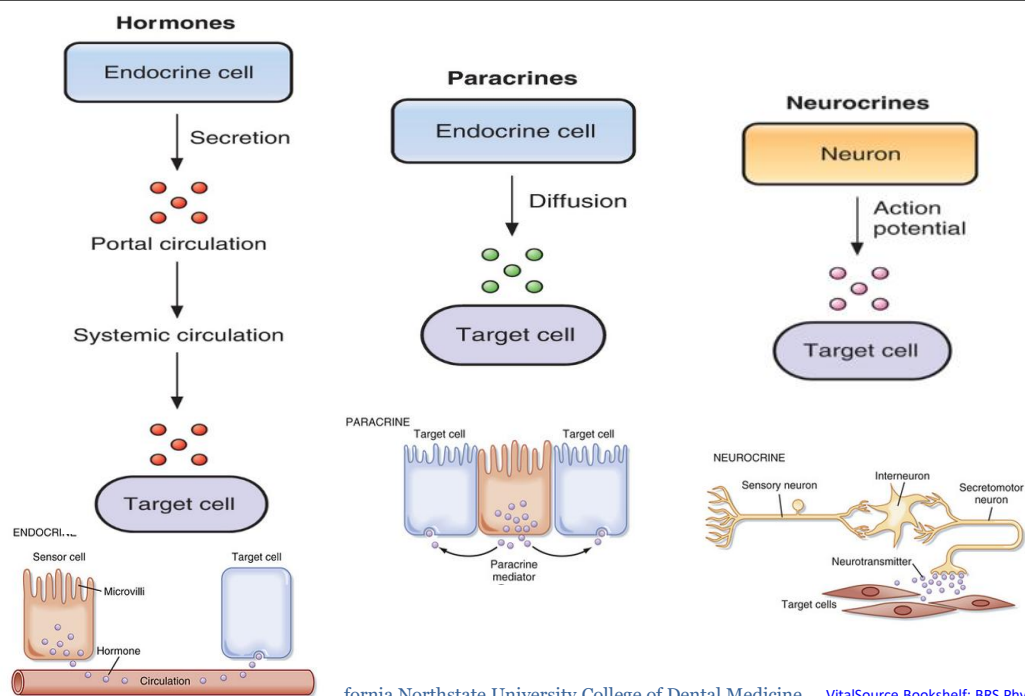


Regulatory Mechanisms in GI system

(Endocrine, paracrine, neurocrine)

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Endocrine Regulation

At least, 15 types of enteroendocrine cells are found in GI mucosa.

1. Gastrin

Endocrine peptide hormone secreted by G enteroendocrine cells of the stomach

Stimuli: stomach distention (especially, amino acids) and vagal stimulation. Gastrin-releasing peptide (GRP, neurotransmitter released from enteric nerve endings).

Inhibitors: Hydrogen ion in the stomach (via a negative feedback mechanism). Somatostatin inhibit gastrin secretion.

Function:

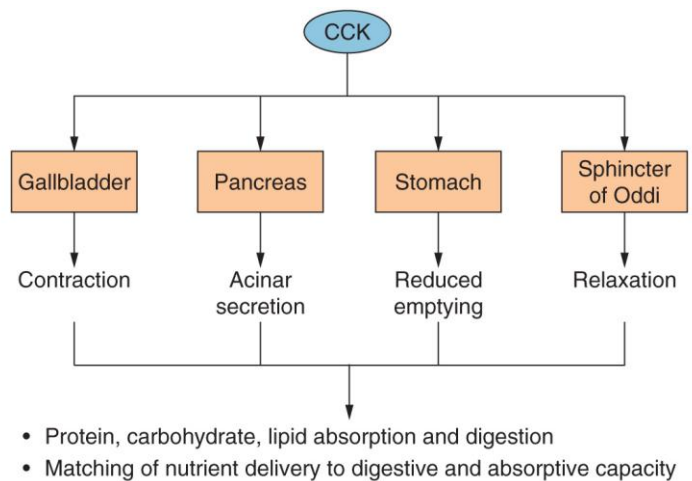
- a. Target parietal cells to secrete hydrogen ion (gastric acid). Chief cells to secrete pepsinogen and ECL cells to secrete histamine.
- b. Trophic action: stimulates growth of gastric and intestinal mucosa.

2. Cholecystokinin:

Peptide hormone secreted by “I” cells of duodenum w endocrine/paracrine action.

Stimuli: fatty acids and monoglycerides, amino acids and peptides.

Functions: Stimulates pancreatic enzyme secretion, pancreatic bicarbonate secretion and contraction of the gallbladder. Inhibits gastric emptying (to allow complete intestinal digestion).



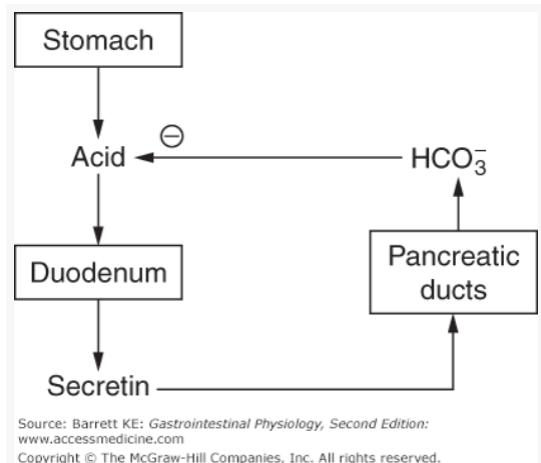
3. Secretin

Small peptide hormone secreted by “S” cells of duodenum.

Stimuli: acidic or low pH and fatty acids in the duodenum.

Functions:

- Stimulates the secretion of bicarbonate and water from the pancreas.
- Increases bile production by the liver.
- Decreases the gastric acid (hydrogen ion) secretion.



4. Glucose-dependent insulinotropic peptide (GIP, aka gastric inhibitory polypeptide)

Endocrine incretin peptide hormone secreted by the intestinal K cells.

Stimuli: in response to fatty acids, amino acids, and oral glucose load.

Function:

- a. Targets β cells and increase stimulates insulin release.
- b. It inhibits hydrogen ion secretion by gastric parietal cells.

5. Other hormones

- a. Motilin, increases GI motility.
- b. Leptin, by adipocytes, decreases appetite. Leptin stimulates anorexigenic neurons and inhibits orexigenic neurons.
- c. Ghrelin, by stomach enteroendocrine cells, stimulates orexigenic neurons and inhibits anorexigenic neurons (increases the appetite).
- Stimulated by simple nutrients (fatty acid, glucose & hydrolyzed proteins)
 - Peptide YY: “intestine L-cells”, inhibition of gastric emptying, pancreatic secretion, gastric acid secretion, intestinal motility, food intake.
 - Glucagon-like peptide-1 (GLP-1), L cells, insulin secretion and epithelial proliferation.

Paracrine regulation

1. Somatostatin

Secreted by D cells of the GI tract and pancreas.

Stimuli: Hydrogen ion in the stomach lumen.

Inhibitors: Vagal stimulation.

Functions:

- a. Inhibits all GI hormones (like, gastrin, secretin, GIP).
- b. Inhibits gastric acid and exocrine pancreas secretions.
- c. Regulates water absorption across the intestinal wall.

2. Histamine

Secreted by mucosal mast and ECL (enterochromaffin) cells of the gastric mucosa. Inhibited by somatostatin.

Functions:

Increases gastric hydrogen ion secretion either directly or via potentiating the effect of gastrin and vagal stimulation.

3. Serotonin

By enteric neurons, mucosal mast cells, and ECL cells.
Regulates smooth muscle function and water absorption across the intestinal wall

Neurocrines

1- Vasoactive Intestinal peptide (VIP)

Secreted from neurons in the GI mucosa and smooth muscle.

- a. Allows the relaxation of the esophagus and proximal part of the stomach to facilitate the passage of food from the esophagus to the stomach
- b. Stimulates pancreatic bicarbonate secretion.
- c. Inhibits gastric hydrogen ion secretion.

2- Enkephalins

Neuropeptides (endogenous opioid). Secreted from nerves in the GI smooth muscle and mucosa. They slow GI motility, inhibit intestinal fluid and electrolyte secretion.

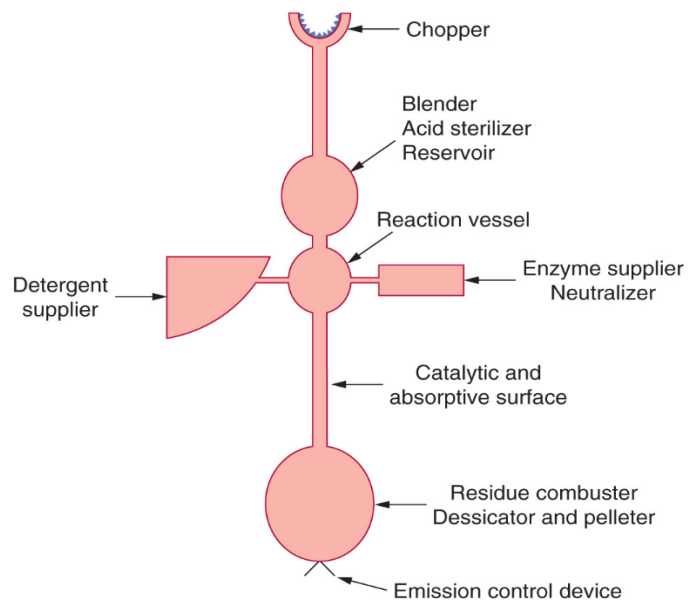
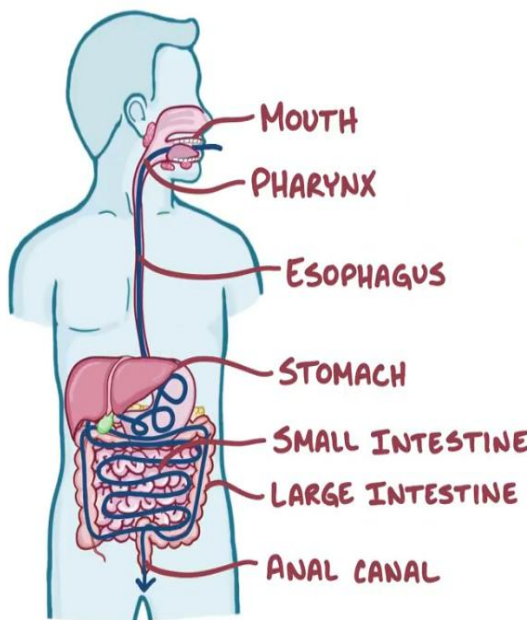
3- Gastrin-releasing peptide (GRP)

GRP is released from the vagus nerve. Stimulates gastrin release from G-cells.

Gastrointestinal System Physiology

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Functions of the GI System

1- Ingestion

- Mastication
- Motility

2- Digestion

- Secretions or hormone release
- Chemical digestion

3- Absorption of digested nutrients into the blood and lymph.

4- Elimination or excretion of indigestible, unabsorbed residues.

5- Host defense (immune response and chemical)

Phases of Digestion

- Cephalic phase
- Oral Phase
- Esophageal phase
- Gastric phase
- Intestinal and colonic phase

Cephalic, Oral and Esophageal Phases

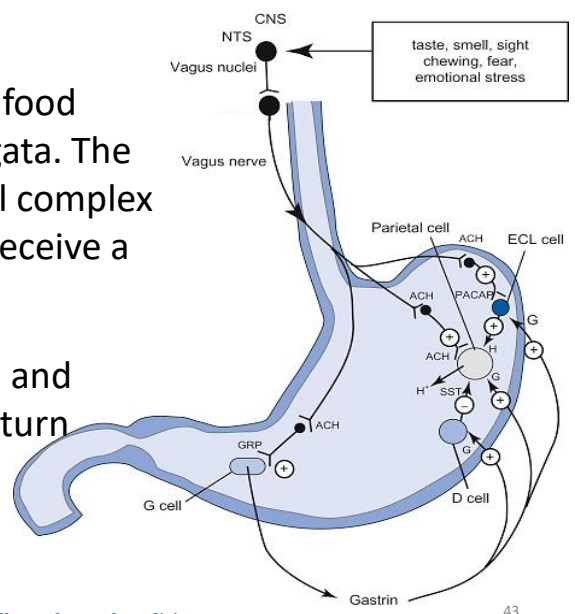
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Cephalic Phase

The taste or smell of food, sensations of food stimulate stimulates the medulla oblongata. The CNS relay information to the dorsal vagal complex to get different parts of the GI ready to receive a meal.

Parasympathetic signaling stimulate ACh and gastrin releasing peptide (GRP) which in turn stimulate gastric secretions and gets the stomach ready for a meal.



Oral Cavity

Functions

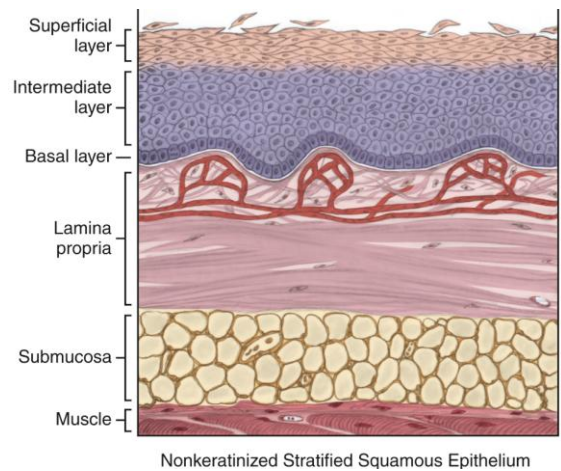
- Provides mechanical processing via the action of the teeth, tongue, and palatal surfaces.
- Provides lubrication by mixing food material with mucus and salivary gland secretions
- Limited digestion of carbohydrates and lipids.
- Absorption. In general, the oral mucosa has insignificant role in absorption, except for the mucosa in the ventral surface of the tongue.

Oral mucosa

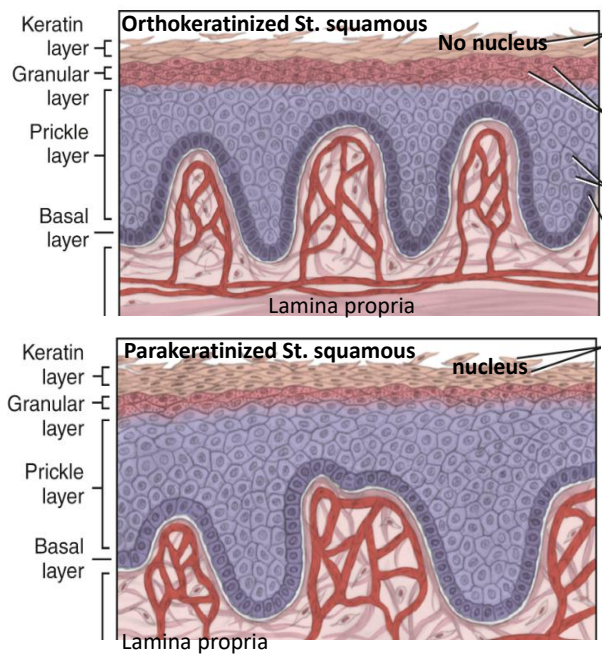
Oral mucosa: stratified squamous epithelium overlying lamina propria, submucosa.

The oral mucosa types:

- **Lining mucosa:** Nonkeratinized stratified squamous epithelial cells. Soft and moist surface (buccal mucosa, labial mucosa, soft palate and ventral surface of the tongue).

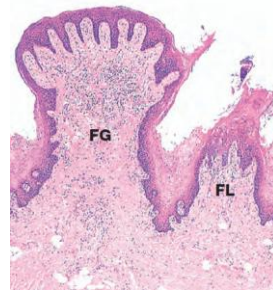


- **Masticatory mucosa:** more resilient tissue texture, can overlay a bone surface.
 - **Orthokeratinized** epithelium, keratin layers has no nucleus, hard palate and attached gingiva.
 - **Parakeratinized**, keratin layers has a nucleus, also in attached gingiva

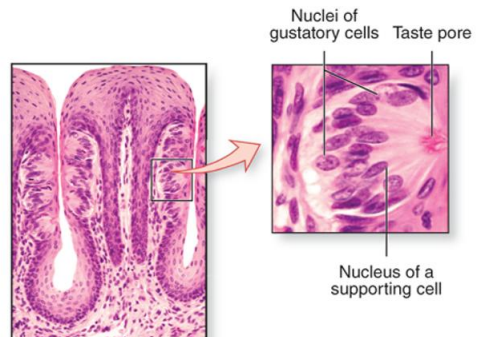


Specialized mucosa: dorsal surface of tongue

- Mucosa is covered by papillae, which contain taste buds.
- Tongue has little or no submucosa.
- Muscle is either intrinsic and extrinsic muscles arranged in bundles.
- Helps with the process of chewing and food maneuver.

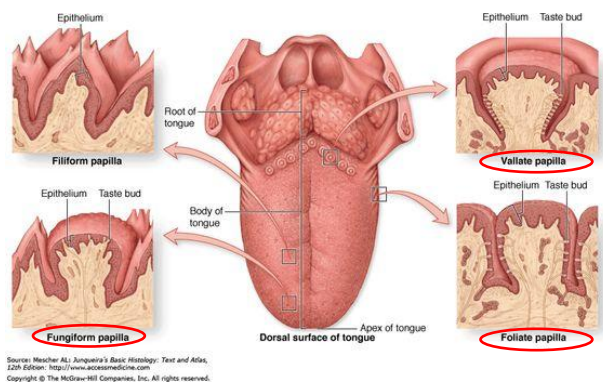


FG=
fungiform
papillae, with
taste buds,
FL=filiform
papillae with
no taste buds

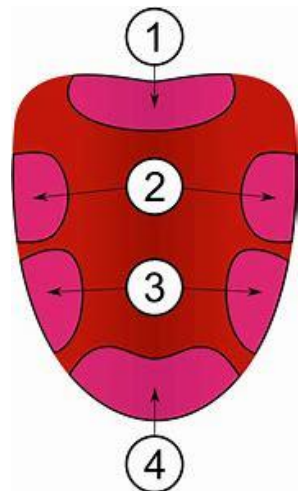
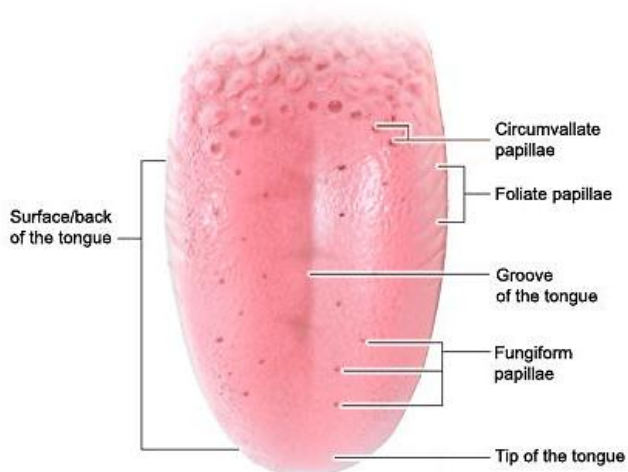


(d) Histology of taste bud
Foliate papillae

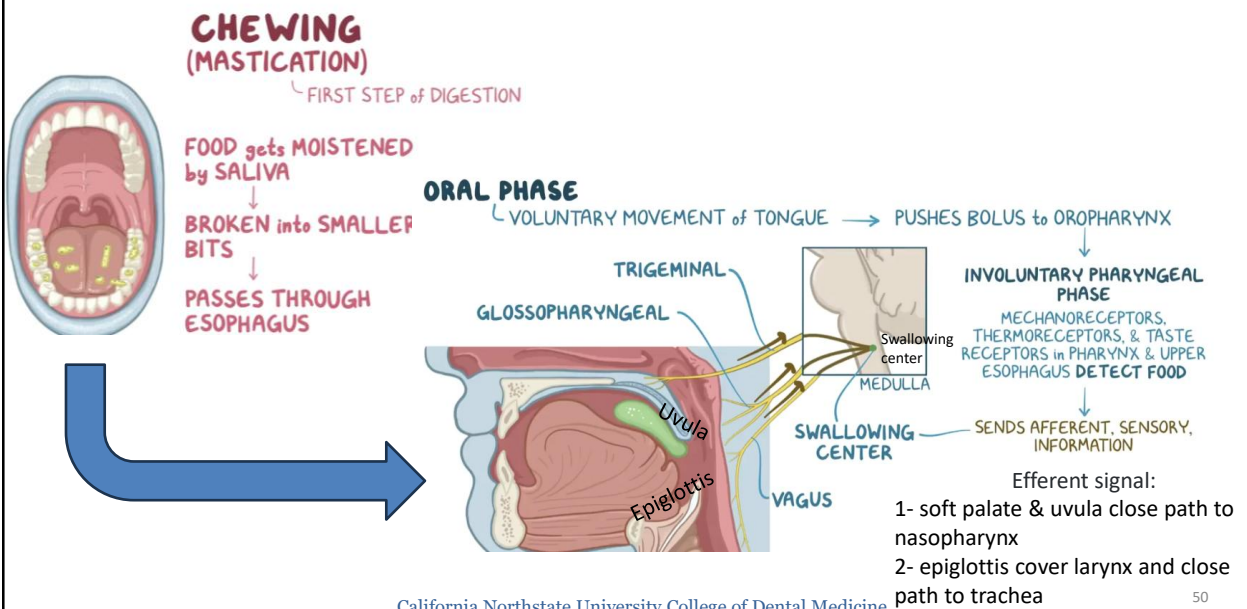
- Circumvallate papillae are most sensitive to bitter tastes.
- Fungiform papillae are most sensitive to sweet and savory (umami) tastes.
- Foliate papillae are most sensitive to salty and sour tastes.
- Filiform papillae are found throughout the entire surface of the dorsal tongue. **No taste buds.** Their rough texture aid in the gripping of food and transfer down to the esophagus, as well as in the cleaning of the mouth and spreading of saliva. Largest population



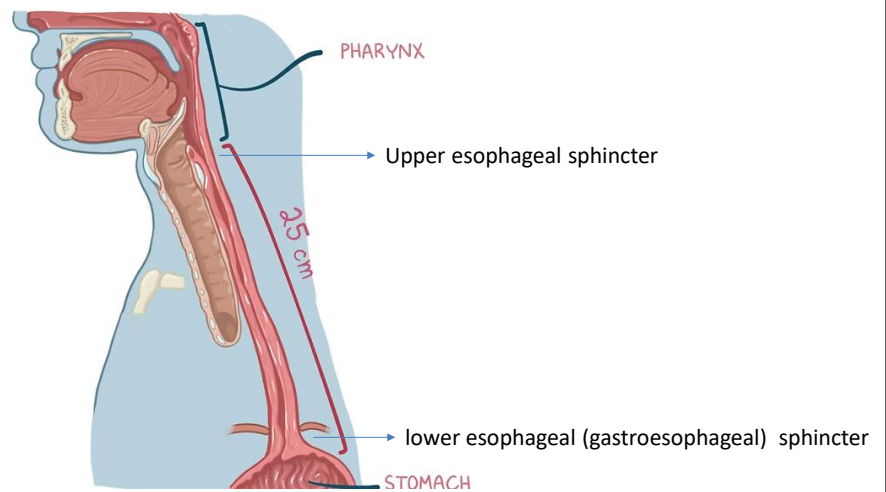
- Match the taste to the location (1-4) (sweet, sour, salty and bitter)



Chewing and Swallowing



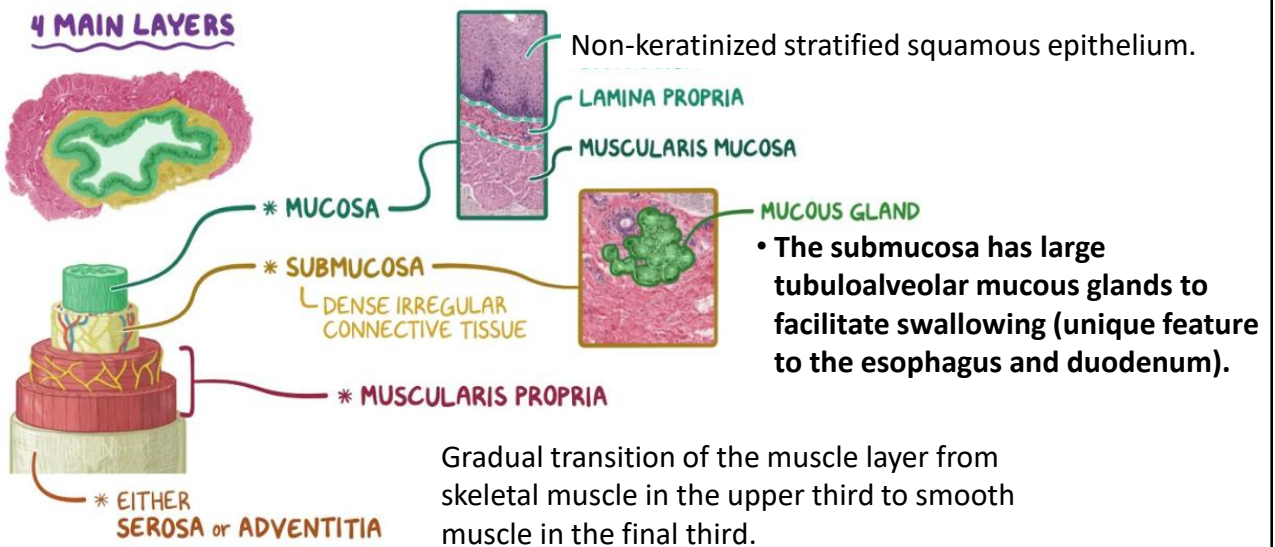
The Esophageal Phase



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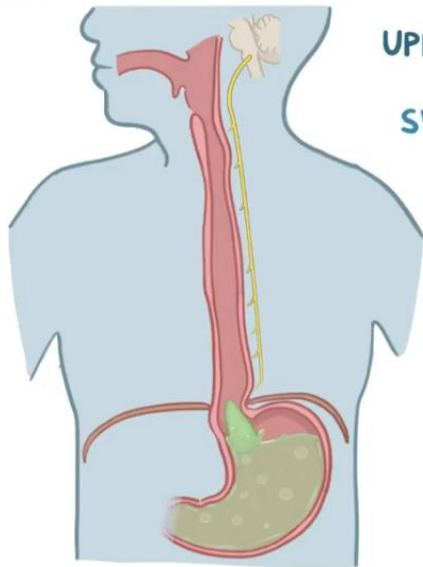
Esophagus Histology



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ESOPHAGEAL PHASE



UPPER ESOPHAGEAL SPHINCTER CLOSES

SWALLOWING CENTER

↳ VAGUS NERVE

↳ INVOLUNTARY MUSCULAR
MOVEMENT of ESOPHAGUS

} PERISTALSIS

LOWER ESOPHAGEAL SPHINCTER

↳ RELAXES

Vagus-mediated via VIP neurotransmitter.
Failure of sphincter can contribute to GERD.

SOLID FOOD — 10 sec.

NON-SOLID FOOD — 1 sec.

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The Gastric Phase

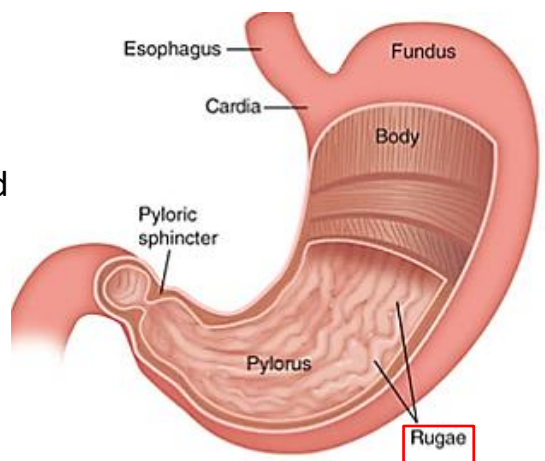
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Stomach

Stomach converts the ingested food bolus into chyme and deliver it to the duodenum.
Chyme= partially digested foods and gastric secretions.

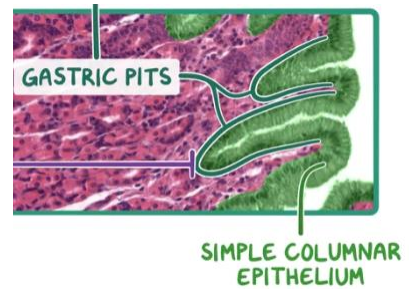
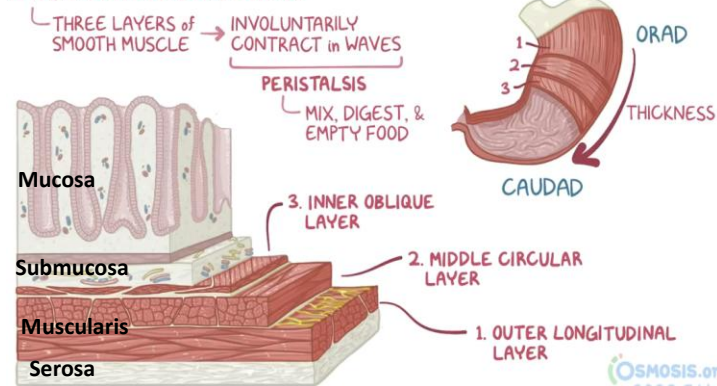
- Rugae are longitudinal gastric folds formed by the mucosal and submucosal layers. Rugae allow the stomach to expand following meal consumption.
- Pyloric Sphincter controls the passage of food to the duodenum.



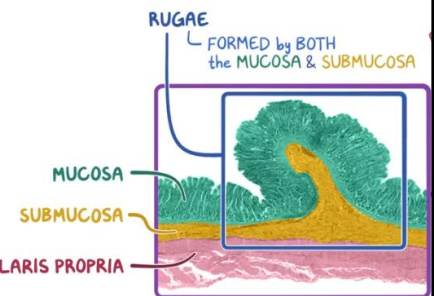
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MUSCULARIS EXTERNA



- Layer thickness differs along the stomach length.
- Muscularis has three smooth muscle layers: an outer longitudinal layer, a middle circular layer, and an innermost oblique layer (unique to stomach).
- Gastric contractions are increased by vagal stimulation & decreased by sympathetic stimulation.



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Functions of the stomach

- Storage, it acts as temporary reservoir for the meal.
- Mechanical and chemical breakdown of food
- Coordinated motor activity to regulate the emptying of its contents into the duodenum.
- Secretion of:
 - Hydrochloric acid to kill ingested microbes and convert pepsinogen to its active form.
 - water for lubrication and to provide aqueous suspension of nutrients.
 - intrinsic factor to absorb vitamin B12 (cobalamin).
 - mucus and bicarbonate to protect the gastric mucosa.

Gastric Motility

1- Receptive relaxation

During the passage of meal from esophagus, the orad region of the stomach (fundus and proximal body) relax to accommodate the ingested meal, mediated by the vagus/enteric nervous system. Vasoactive intestinal peptide (VIP) relaxes the smooth muscles of esophagus and orad regions.

2- Mixing and digestion

Peristalsis help the stomach mix, digest and push food towards the duodenum. The caudad region of the stomach contracts to mix the food with gastric secretions.

3- Gastric emptying

- The caudad region of the stomach contracts to propel food through the pyloric sphincter & into the duodenum. Thickness of muscularis externa increases towards the distal part of the body and pylorus (collectively termed caudad).
- Gastric emptying usually occur after few hours of receptive relaxation.
- The rate is usually affected by the type and amount of ingested food. For example, fatty meals and acidic food tend to slow down the gastric emptying by stimulating the release of CCK.

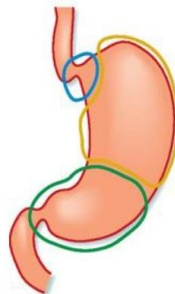
Gastric glands

- Gastric glands make up around 2-3 liters of secretions everyday, with pH 1-3.5. Stomach mucosa has of 2 types of tubular glands in the gastric pits (pyloric and oxyntic):

1- Pyloric glands (pylorus of stomach, minor population).

Pyloric glands secrete mucus.

Functions: lubricates the stomach wall and protects the gastric mucosa from acidic secretions.



Region	Luminal secretion	Motility
LES* and cardia	Mucus HCO_3^-	Prevention of reflux Entry of food Regulation of belching
*LES is part of the esophagus		
Fundus and body	H^+ Intrinsic factor Mucus HCO_3^- Pepsinogens Lipase	Reservoir Tonic force during emptying
Antrum and pylorus	Mucus HCO_3^-	Mixing Grinding Sieving Regulation of emptying

[Digestive Tract | Junqueira's Basic Histology Text and Atlas, 16e](#)
[| AccessMedicine | McGraw Hill Medical \(cnsu.edu\)](#)

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2- oxyntic (aka, gastric) glands (occupy the proximal 80% of the stomach in gastric pits).

Several types of cells:

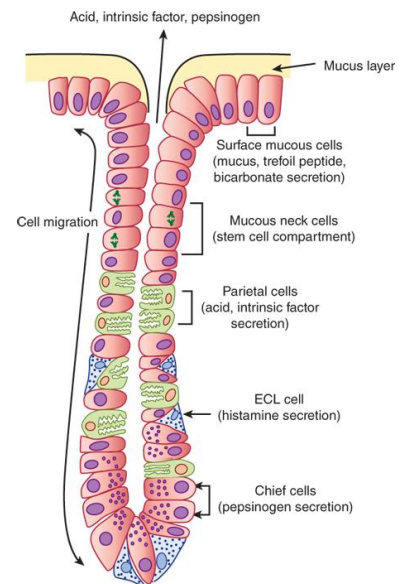
a- Mucous neck cells, line the neck of the pit.

Secrete mucus & bicarbonate to protect the gastric epithelia from the acidity of other gastric secretions.

b- Parietal (aka, oxyntic) cells, gastric fundus.

- **Hydrochloric acid (HCl)** in response to Ach, gastrin or histamine stimulations. Kills food-derived bacteria and facilitate digestion/absorption.
- **Intrinsic factor** necessary for vitamin **B₁₂** uptake.

Damage of parietal cells can lead to pernicious anemia.



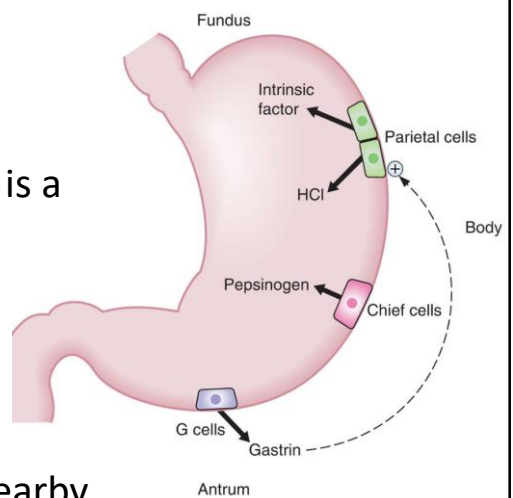
Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan: Ganong's Review of Medical Physiology, Twenty-Sixth Edition Copyright © McGraw-Hill Education. All rights reserved.

Gastric (oxyntic) gland from the body of the stomach
ECL, Enterochromaffin-like cells.

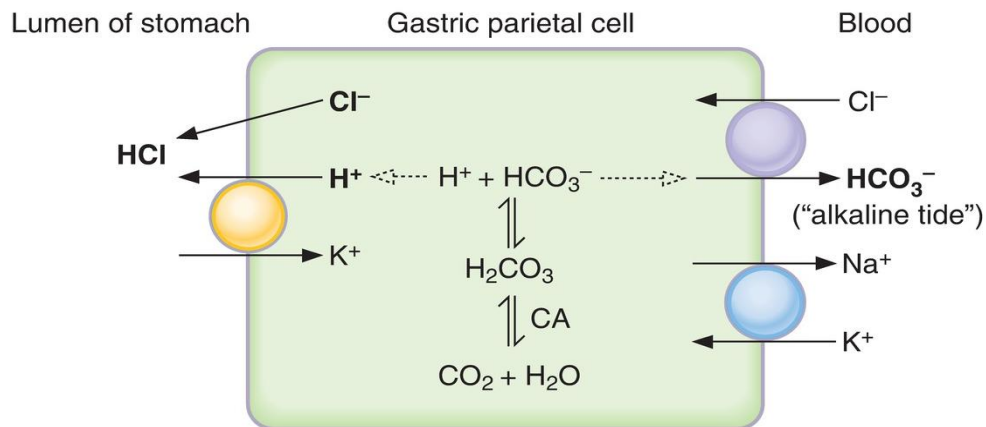
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c- Chief cells in the fundus. For secreting pepsinogen. HCl converts the inactive “pepsinogen” to the active “pepsin. Pepsin is a proteolytic enzyme responsible for protein digestion (secreted in response to vagal stimulation).

d- G cells in pyloric region. Secrete gastrin in the pylorus to regulate nearby tissues during food digestion. Stimulates the secretion of HCl via activating cholecystokinin B receptor in the parietal cells.



Mechanism of Hydrochloric Acid Secretion



Parietal cells: Hydrogen ion is secreted in the stomach lumen in exchange of potassium ion (**hydrogen-potassium pump**; H⁺/K⁺ ATPase). Bicarbonate is exchanged with chloride ion from the blood & chloride is secreted into the stomach lumen.

Alkaline Tide & Metabolic Alkalosis

- To enable HCl secretion, bicarbonate is exchanged with chloride ion between blood and stomach.
- Increased bicarbonate in blood results in an increase of the blood pH (“alkaline tide”). To maintain pH balance, bicarbonate is secreted with pancreatic secretions to neutralize chyme in the duodenum.
- Vomiting can cause metabolic alkalosis, how?
 - Gastric contents will not go to the small intestine which prevent the pancreatic bicarbonate secretion. Therefore, blood becomes more alkaline leading **to metabolic alkalosis**.

Regulation of HCl secretions

Stimulation

Vagal stimulation (direct and indirect), Gastrin (via interaction with cholecystinin B receptor on the parietal cells) and histamine via H₂ receptors.

Inhibition

Physiological: Low pH (<3, negative feedback mechanism), somatostatin (major inhibitory mechanism) and prostaglandins (inhibit hydrogen ion secretion, stimulate bicarbonate and maintain mucosal barrier).

Pharmacological: inhibit the H⁺,K⁺-ATPase pump, antagonist for H₂ receptor.

Gastroduodenal Mucosal Defense

- 1- Layer of mucus and bicarbonate
 - 2- Surface epithelial cells tight junction and their renewal rate (every 3-5 days)
 - 3- Microcirculation
 - 4- Prostaglandin (like PGE_2): decrease gastric H^+ secretion
- Maintain the mucosal barrier and stimulate HCO_3^- secretion.

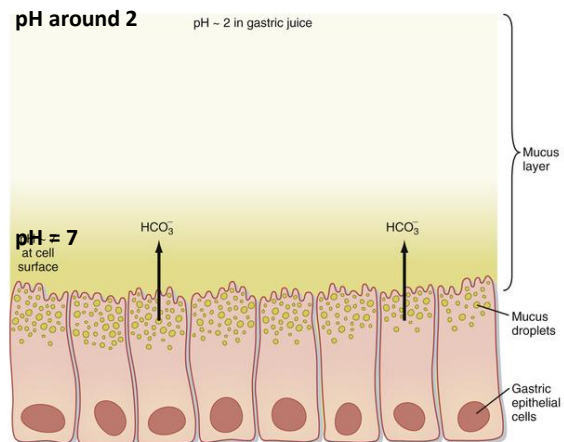


FIG. 29.11 The surface of the stomach is protected by the gastric mucosal barrier. Buffering by the HCO_3^- -rich secretions and the high viscosity of the layer of mucus allow the pH at the cell surface to remain near 7, whereas the pH in the gastric juice in the lumen is 2.

Esophageal and Gastric disorders Gastritis, and PUD

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Gastritis

Inflammation of stomach lining. Symptoms: range from asymptomatic, to epigastric pain, nausea, early satiety .

- **Acute gastritis:** Imbalance of mucosal defense and rate of acid production. Causes: excessive alcohol use, smoking or chronic NSAIDs. Usual symptoms: epigastric pain, nausea, and vomiting. Can progress to ulcer or GI bleeding. Management: manage trigger and avoid excessive NSAIDs use.
- **Chronic gastritis:** infection with *H. pylori* (most common) or autoimmune gastritis can cause chronic ongoing inflammation leading to gastric atrophy. Complications: can progress to an ulcer or gastric adenocarcinoma. Management: manage trigger like *H. pylori* with antibiotics and avoid excessive NSAIDs use.

Peptic Ulcer Diseases

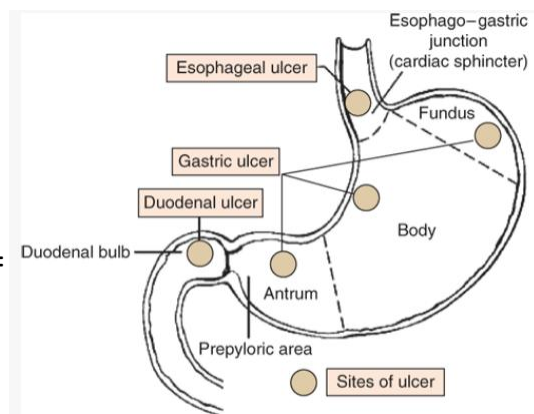
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- Damage in the mucosal surface of the stomach or duodenum that extends through the muscularis mucosa into the submucosa or deeper layers.
- Percent of adults with diagnosed ulcers: 5.9% (14 million).
- Slightly higher prevalence in males (M/F = 3:1 for DU and 1.5-2:1 GU).

Risk factors:

- 1- Infection with *H.pylori*.
- 2- Excessive NSAIDs use.
- 3- Lifestyle (alcohol consumption, stress, coffee and smoking).



Source: J.B. Halter, J.G. Ouslander, S. Studenski, K.P. High, S. Asthana, M.A. Supiano, C. S. Ritchie, K. Schmadier, W.R. Hazzard, N.F. Woolard: Hazzard's Geriatric Medicine and Gerontology, 8e: Copyright © McGraw Hill. All rights reserved.



Gastric Ulcers
Pain greater with meals

Less common, develop later in life

Etiology

H. pylori - 70%
 NSAIDs

Mechanism

 Decreased mucosal protection against stomach acid

Risk of gastric carcinoma

 Biopsy margins to rule out malignancy



Duodenal Ulcers
Pain relieved with meals

More common

Etiology:

H. pylori - 90%
 Zollinger–Ellison syndrome

Mechanism:

 Increased gastric acid secretion
 Decreased mucosal protection

Not associated with malignancy


Elevated gastric acid secretion or decrease mucosal defense mechanism.

Pathophysiology

- Imbalance between the GI erosive factors and GI protective mechanisms (mucus, bicarb, PGE_2) leading to the exposure of the GI mucosa to gastric acid and pepsin.
- Alcohol and NSAIDs can directly injure the gastric mucosa. Alcohol alters cell permeability and can cause cell death. NSAIDs disrupt mucosal resistance by impairing prostaglandin production.
- *H. pylori* causes inflammation of the gastric mucosa by producing proteases that can induce mucosal damage.

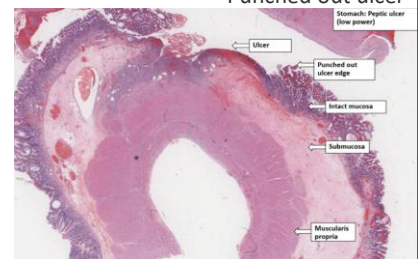
GASTRIC ULCERS



SUBMUCOSA



Punched out ulcer



<https://medicine.nus.edu.sg/pathweb/virtual-pathology-museum/00586-2/>

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Symptoms

- Epigastric pain (with food in GU and 2-3h after food in DU and relieved by eating)
- Nausea, vomiting and bloating.
- Early satiety.
- Dyspepsia or feeling uncomfortably full.

Complications:

Bleeding in the GI tract presented as bloody stools (melena) or hematemesis (relatively common complication), duodenal perforation or malignancy (less common).

Diagnoses

- Endoscopy for diagnosing gastric and duodenal ulcers.
- Test for *H pylori* infection via stool antigen detection or urease breath test.

Management

- 1- Pharmacological agents to suppress the gastric acid production.
- 2- Treatment of *H. pylori*.
- 3- Use of cytoprotective agents as bismuth subsalicylate.
- 4- Stop NSAID use and other GI irritants.

Zollinger-Ellison Syndrome (Gastrinoma)

Cause: Gastrin-secreting tumor.

Etiology: Gastrin-secreting neuroendocrine tumors (gastrinoma) results in increased gastrin secretion. Excess gastrin stimulates hydrochloric acid secretion from the parietal cells.

This syndrome is associated with multiple peptic ulcerations in the stomach and duodenum (uncommon syndrome).

Resources

